Team 6

- **Separating Lower Risk from Timing Skill:** The timing strategy had volatility of approximately 11%, compared with 15.9% for buy-and-hold, and its drawdown advantage also diminished when ex-post leverage was used to equalize risk. Would the stabilization of risk and reduction in drawdown persist against a simple stock–cash allocation rule that uses only past information, without looking ahead?
- **Consistency Between Training and Trading Rules:** Dec increased the Sharpe ratio for four of six factors, but improved its own utility objective for only two. Your team explained that c is selected once per fit during training but reset annually during evaluation, and that monthly forecasts are derived by extending daily predictions. If these differences were removed and training and trading rules aligned, would the conclusion that Dec has only limited incremental value still hold?
- **Validation Independent of Design Exploration:** The 1998 start date, weight caps, and comparison rules were determined during exploration of the same dataset, and the multiple-testing correction did not cover the entire search. If the design were fixed before evaluation on a separate period or market, would the finding that "risk becomes more stable but Sharpe-ratio gains are not significant" still hold?
- Why did you run the main test on the Ken French data (1963–2025) rather than the course CRSP file (2000–2020)? Over 2000–2020 alone, the timed portfolios have a Sharpe of 0.53–0.57 against 0.435 for the market. Is that difference significant?
- Exposure is above 1 in 41–45% of months, and borrowing is assumed at the riskfree rate with no spread. If leverage were not allowed (cap = 1.0), how would the drawdown reduction and the Sharpe results change?
- You showed that recovery months return 1.59% a month while the rule holds only 0.35 of the market. Yet holding at least full exposure in those months lowers the Sharpe from 0.61 to 0.59. How do you reconcile these two results?
- Why did directly optimizing investment utility fail to produce consistent out-of-sample improvements? Which explanation do you consider most likely: overfitting, training instability, or changing market conditions?
- Does the risk reduction remain when compared with a fixed-allocation strategy with the same average stock exposure? This would help distinguish the effect of holding fewer stocks overall from the effect of reducing exposure at risky times.

- For the equal-weighted stock benchmark, how were stocks with missing prices handled each month? Did you check how sensitive the results were to that treatment?
- The source data goes back to 1963 - why specifically start the test window in 1998? Curious whether this was to lock in a window that happened to show at least some significant results. (worried this sounds too aggressive)
- There is no sensitivity analysis on the 1.5x exposure cap or the target volatility value itself - how would conclusions change if these numbers were different?
- You timed each of the six factors separately. My understanding is real portfolio management typically allocates risk budgets using the full covariance structure across factors. You didn't use cross-factor covariance, treating each factor as an independent single asset - is there a reason for not using this covariance-based design?
- Regarding Q2: How similar is the ranking by forecast accuracy to the ranking by Sharpe ratio? Did you calculate it?
- Regarding Q3: The way c was chosen differs between training and evaluation. Could that difference have affected the results? Which of $p = 0.39$ and $p = 0.048$ is the value before the correction?
- Data conditions: If GARCH-LSTM were also trained on data starting from 1963, would the results be the same?
- Exposure is set based on the inverse of the forecast variance, with a cap of 1.5. Because of the cap, two different forecasts can end up giving the same position. Apart from overall forecast error, does the accuracy ranking of the forecasts (RV, GARCH, GJR-GARCH, GARCH-LSTM and VIX2) stay the same if you only look at months where forecast errors materially change the position?
- The drop in volatility fluctuations was significant for the GARCH models, but the gain in certainty-equivalent return wasn't. Steadier risk is worth more to some investors than others. What kind of investor, or what risk limits or goals, would make this stability a useful improvement in practice?
- The GARCH-LSTM trained on investor utility improved the Sharpe ratio for four of the six factors, but its own utility objective for only two. Since these are different measures, which one do you use to judge whether the model worked? Also, was the risk-aversion coefficient chosen in advance for a target investor, or picked by comparing values on a separate validation period?
- **Risk Reduction and Portfolio Performance**: As you see from the results, volatility

timing reduced the maximum drawdown from around −50.3% with the buy-and-hold strategy to between −21.3% and −26.7% for the timed portfolio strategies. The improvement in the Sharpe ratio, however, was not statistically significant. If doing so doesn't significantly improve risk-adjusted investment returns, do you believe it is enough to render this strategy appealing to those looking to invest?

- **Forecast Accuracy**: In your presentation, you identified that the GARCH-LSTM model was the most accurate forecaster of the three GARCH models, but its performance in the portfolio was not as strong as the other two models. What do you think is the reason why a more accurate volatility prediction did not result in a higher Sharpe ratio?
- **Improving the Timing Strategy:** The volatility timings strategy was better in all the financial crises periods but worse in the periods of sudden market declines and fast recovery periods. Would you think that if you used more frequent trading signals like daily or weekly signals, as opposed to monthly signals, the strategy would be better, and what other potential risks and trading costs would be involved?